A Web-Based Multi-Branch Music Academy Management System with Smart Attendance and Performance Management for Father and Sons Music Lessons

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February 2026

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A **Capstone Project** Presented to the

Faculty of the College of Information Technology

Cagayan de Oro College

Cagayan de Oro City

In Partial Fulfillment

of the Requirements for the Degree

Bachelor of Science in Information Technologies

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CHAPTER 1
THE PROBLEM

Introduction

Music academies play an important role in developing student's musical skills, discipline, and performance abilities through structured lessons and regular practice. In Cagayan de Oro City, Father and Sons Music Lessons is a multi-branch music academy that provides music education to students through its SM Uptown and Downtown branches. As the number of enrolled students, instructors, and learning activities increases, effective management of lessons, attendance, and student performance becomes essential to ensure smooth operations and consistent quality of instruction across branches.

At present, Father and Sons Music Lessons utilizes a combination of an existing system, paper-based records, and spreadsheet files to manage its operations. However, the current setup lacks essential features needed to fully support the academy's growing operational requirements. Attendance monitoring and performance tracking are primarily handled through manual attendance sheets and Excel-based records, while some student and session information is still documented on paper. This fragmented approach often results in delayed report generation, inconsistent data between branches, data redundancy, and an increased risk of errors or data loss. As a result, administrators and instructors experience difficulty in efficiently monitoring student progress, attendance compliance, and overall academy performance.

With the continuous advancement of web technologies, many educational institutions have shifted toward centralized and automated systems to improve data accuracy, accessibility, and operational efficiency. Web-based management systems enable real-time access to

information, secure storage of records, and improved coordination across multiple locations. However, many existing systems used by music academies focus only on basic record-keeping and do not sufficiently integrate attendance monitoring and performance management into a unified platform.

In response to these challenges, this study proposes the development of a Web-Based Multi-Branch Music Academy Management System for Father and Sons Music Lessons with Smart Attendance and Performance Management. The proposed system aims to centralize lesson management, automate attendance monitoring, and systematically track student performance across branches. By integrating these processes into a single web-based platform, the system is expected to enhance operational efficiency, improve the accuracy of records, and support better monitoring and decision-making for academy administrators and staff.

Statement of the Problem

The current management process at Father and Sons Music Lessons relies on a combination of an existing system, paper-based records, and Excel spreadsheets to handle student attendance, lesson schedules, and other academy operations. These traditional and partially digital methods slow down daily processes, limit system functionality, and make it difficult to access real-time and consolidated data across multiple branches, which negatively affects operational efficiency and academic monitoring.

Specific Problems:

1. Fragmented record-keeping

Student information, attendance, and lesson schedules are stored in paper and spreadsheets, causing redundancy and inconsistency.

2. Inefficient and Vulnerable Attendance Tracking

Manual attendance processes at Father and Sons Music Lessons make it difficult to monitor student attendance accurately and in real-time, and physical attendance cards are prone to manipulation by students.

3. Limited Performance Monitoring and Analytics

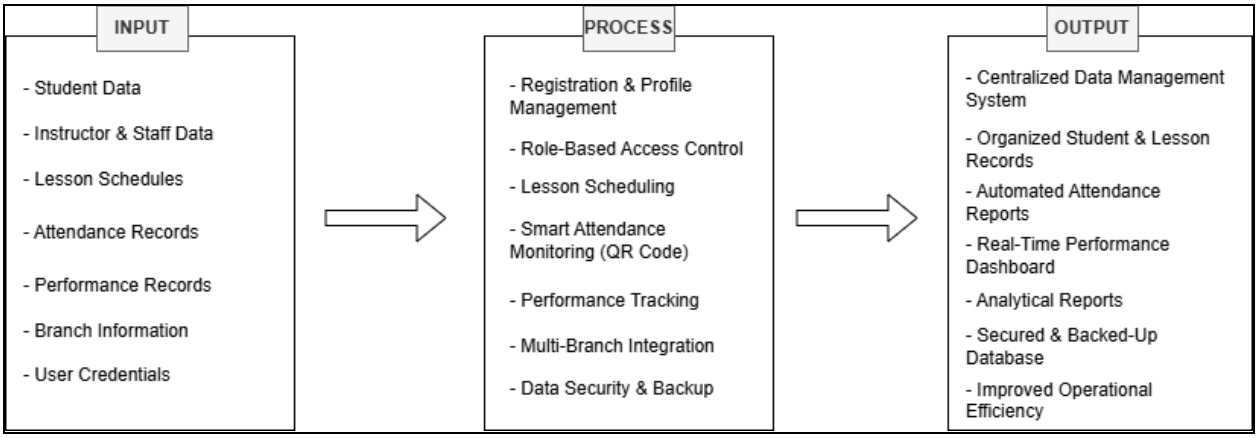
The academy lacks a centralized system or dashboard to track student progress, lesson completion, and instrument performance, making it difficult for instructors and administrators to monitor and support students effectively.

4. Inefficient Report Generation

Attendance, performance, and lesson reports are generated slowly due to manual and fragmented data sources.

Conceptual Framework (IPO Model)

This study is anchored on the Input–Process–Output (IPO) Model, which illustrates how data inputs are transformed through defined system processes to generate meaningful outputs that support effective management and decision-making in the proposed Web-Based Multi-Branch Music Academy Management System.



Objectives of the Study

General Objective:

The general objective of this study is to design, develop, and evaluate a secure and scalable Web-Based Multi-Branch Music Academy Management System with Smart QR Code-Based Attendance and Performance Monitoring for Father and Sons Music Lessons, aimed at centralizing data management, automating attendance validation, streamlining lesson scheduling, and improving student performance tracking across multiple branches.

Specific Objectives:

Specifically, this study aims to:

1. **Develop** a web-based system that centralizes music lessons, student profiles, and performance records across multiple branches.
2. **Implement** a QR code–based attendance system that automates attendance validation and ensures accurate and reliable attendance records.
3. **Generate** automated attendance and student performance reports to improve monitoring and record management.

4. **Provide** real-time and consolidated system data that supports administrative decision-making for effective academy management.
5. **Design** a user-friendly interface that allows instructors and administrators to manage lessons, attendance, and student performance efficiently across all branches.

Significance of the Study

This study will be beneficial to the following stakeholders:

Students – The system allows students to access organized lesson records and monitored performance data, enabling them to track their progress more effectively.

Instructors – The system assists instructors in accurately monitoring attendance and student performance while reducing manual record-keeping tasks.

Desk Staff – Automated attendance and record management lessen workload and improve efficiency in handling daily administrative operations.

School Management – The system provides school management with centralized, real-time access to operational and academic data across branches, supporting planning, supervision, and policy implementation.

Administrator and Branch managers – The system enables administrator and branch managers to manage records, generate reports, and monitor academy performance across multiple branches, supporting informed decision-making.

Parents/Guardians – The system allows parents and guardians to monitor student attendance, lesson schedules, and performance updates, promoting transparency and engagement in the students' learning progress.

Researchers (The Research Group) – The study enhances the researchers' knowledge and skills in system analysis, design, and development, and serves as practical experience in developing a web-based, multi-branch educational management system.

Future Researchers – This study may serve as a reference for future research related to web-based, multi-branch educational management systems and smart attendance solutions.

Scope and Limitations

This study focuses on the development of a web-based management system for Father and Sons Music Lessons, which will be accessible through both desktop and mobile web browsers. The system will cover multi-branch operations, specifically the Uptown and Downtown branches in Cagayan de Oro City, and will provide core functionalities such as music lesson management, smart QR code-based attendance monitoring, student performance tracking, and automated reporting. User roles within the system will include Admin, Branch Manager, Instructor, and Desk Staff ensuring role-based access and management.

However, the system will be implemented exclusively for Father and Sons Music Lessons during its development and pilot testing phase. Attendance monitoring relies exclusively on QR code technology, without using advanced biometric or facial recognition methods. The system

focuses on a web-based platform and does not include a dedicated mobile application. These delimitations ensure the study remains focused on developing a functional, secure, and user-friendly management system for multi-branch music academy operations.

Definition of Terms

Web-Based System

Conceptual: A system that is accessed through a web browser using an internet connection.

Operational: The online platform developed to manage music academy operations, lessons, attendance, and performance records for Father and Sons Music Lessons.

Music Academy Management System

Conceptual: A system designed to manage academic and administrative activities of a music academy.

Operational: The proposed system that handles lesson scheduling, student records, attendance, and performance tracking across multiple branches.

Smart Attendance

Conceptual: A technology-assisted method of recording attendance automatically or digitally.

Operational: The system feature used to record and monitor student attendance in real time across branches.

Performance Management

Conceptual: The process of monitoring and evaluating individual progress and outcomes.

Operational: The system module that tracks students' musical progress and performance records.

Multi-Branch Management

Conceptual: The coordination and control of operations across multiple locations.

Operational: The system's ability to manage data and operations for both Uptown and Downtown branches in a centralized platform.

Data Security

Conceptual: The protection of digital data from unauthorized access or loss.

Operational: The login controls, role-based access, and data protection features implemented in the system to secure academy information.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the review of related literature and studies that establish the theoretical and empirical foundations of the present research on web-based music academy management systems. It includes local and foreign literature and studies relevant to music education management, attendance monitoring systems, QR code–based attendance solutions, performance evaluation systems, and web-based information systems. The chapter also provides a synthesis of the reviewed works and identifies research gaps related to the centralized management of lessons, attendance, and student performance in multi-branch music academies, which the current study seeks to address.

Local Literature

Recent Philippine literature shows a growing adoption of QR code–based systems for attendance monitoring and educational management. Locally developed systems focus on digitizing administrative processes such as student registration, scheduling, record keeping, and attendance tracking. These innovations aim to reduce manual documentation, minimize clerical errors, and improve institutional efficiency.

Several studies highlight the effectiveness of QR code–based attendance systems in educational settings. For instance, an employee attendance and management system at Sorsogon State University, Castilla Campus, demonstrated improvements in accuracy, data retrieval, and report generation (Agripa & Astillero, 2022). Similarly, the use of QR code–enabled ID cards for student attendance monitoring at Jesus F. Magsaysay Technical Vocational High School was shown to enhance punctuality and attendance compliance (Pantilon, 2022). Other systems

integrate additional features, such as web-based monitoring combined with GSM modules for SMS notifications, allowing parents to receive real-time updates on student attendance (Bondoc, 2022). Animated QR codes have also been explored to ensure legitimacy and prevent fraud, further improving accuracy and reliability (Saad & Zidan, 2022).

Local Studies

Empirical research in the Philippines confirms that QR code–based attendance systems are effective and well-received. According to Agripa and Astillero (2022), automated attendance reduces human error and streamlines reporting. According to Pantilon (2022), QR code–enabled attendance significantly improves punctuality and reliability. Both studies indicate that QR systems are perceived as user-friendly, accurate, and efficient compared to manual methods.

According to Wanjare et al. (2024), a contactless smart attendance system combining facial recognition with QR codes improves security and operational efficiency by allowing accurate verification while minimizing physical contact — a critical innovation in the post-pandemic educational context. According to Mamore et al. (2022), the WeLink system, a web-based management platform, effectively tracks attendance, monitors student progress, and manages payments for small tutorial centers.

However, local studies still largely focus on single-function attendance monitoring. According to Saad and Zidan (2022), multi-feature systems that integrate lesson management, instructor evaluation, and performance tracking remain underexplored in Philippine music education institutions. According to Bondoc (2022), fully integrated platforms combining multiple modules are also limited in practice, highlighting a gap in research on comprehensive management systems for music academies.

Foreign Literature

International research highlights the development of advanced QR code–based and intelligent attendance systems. According to Saad and Zidan (2022), security mechanisms such as dynamic or animated QR codes help prevent fraudulent check-ins and improve data integrity. According to Hu and Yang (2022), attendance systems can be integrated within broader academic management platforms, where attendance data contributes to curriculum evaluation, performance analytics, and institutional decision-making. In music education, performance evaluation models are widely applied to assess instructional and institutional quality. According to Wang (2022a), multi-criteria decision frameworks provide data-driven support for strategic planning, while Wang (2022b) reports that the Analytic Hierarchy Process (AHP) can be used to evaluate institutional and instructional quality. According to Wijanarko and Mailoa (2022), a web-based scheduling system for music courses enhances lesson management and improves overall institutional efficiency.

Despite these advancements, most foreign systems remain conceptual, experimental, or limited to single institutions. According to Hu and Yang (2022), multi-branch private music academies are rarely the primary focus. According to Wang (2022b), fully integrated platforms combining attendance, scheduling, instructor management, and performance tracking remain limited in practice.

Foreign Studies

Empirical studies abroad confirm that intelligent management systems enhance educational administration. According to Hu and Yang (2022), dynamic QR code systems and other automated attendance solutions reduce fraud and improve reliability. According to Wang (2022a), structured performance evaluation models have been validated as effective tools for institutional monitoring.

Research in music education shows that integrated systems combining attendance, scheduling, and evaluation modules improve coordination between instructors, students, and administrators, supporting timely reporting and performance monitoring. According to Wang (2022b), such systems enhance instructional and institutional efficiency. According to Wijanarko and Mailoa (2022), web-based scheduling platforms improve lesson management in music courses. According to Ujakpa et al. (2022), a School Management Information System (SMIS) facilitates online student registration, academic record management, and fee tracking, demonstrating the benefits of centralized platforms. However, many of these implementations focus on single-function modules, either attendance or evaluation, rather than fully integrated multi-branch platforms.

Synthesis of Related Literature and Studies

The reviewed literature and studies collectively demonstrate that QR code and contactless attendance technologies, alongside web-based platforms, significantly enhance the efficiency, accuracy, and reliability of educational management processes. Both local and foreign research confirm that digitized attendance monitoring reduces clerical errors, minimizes fraudulent check-ins, enhances real-time data access, and promotes transparency within institutions (Agripa

& Astillero, 2022; Pantilon, 2022; Bondoc, 2022; Saad & Zidan, 2022). Moreover, combining biometric technologies, such as facial recognition, with QR code attendance further improves security and operational efficiency, particularly in post-pandemic learning environments (Wanjare et al., 2024).

Structured performance evaluation frameworks, as highlighted in foreign studies, provide systematic approaches for institutional monitoring, data-driven decision-making, and strategic planning (Wang, 2022a; Wang, 2022b). Web-based platforms, such as tutorial management systems and school information systems, further support administrative efficiency, enabling attendance tracking, lesson scheduling, progress monitoring, and centralized recordkeeping (WeLink, 2022; Hu & Yang, 2022; Wijanarko & Mailoa, 2022; Ujakpa et al., 2022). These findings collectively establish a strong technological and empirical foundation for developing integrated educational management systems in specialized learning environments, including music academies.

However, critical analysis of the literature reveals several consistent limitations. First, most existing systems remain function-specific, focusing primarily on attendance monitoring or academic record management rather than offering comprehensive administrative solutions (Agripa & Astillero, 2022; Pantilon, 2022; Bondoc, 2022). Second, implementations are largely limited to single-campus or single-location institutions, lacking the centralized control mechanisms necessary for multi-branch operations (Hu & Yang, 2022; Wijanarko & Mailoa, 2022; Ujakpa et al., 2022). Third, performance monitoring in existing systems emphasizes general academic metrics and does not adequately address skill-based or music-specific performance evaluation (Wang, 2022b; Hu & Yang, 2022). Fourth, even when QR code–based or facial recognition attendance systems are implemented (e.g., Wanjare et al., 2024; Saad & Zidan,

2022), they are seldom integrated with broader administrative modules such as lesson scheduling, instructor assignment, and structured performance tracking.

These observations indicate that while technological components like QR code attendance, animated QR codes, facial recognition, and web-based databases are well-established, their integration into a unified, music-specific, multi-branch management platform remains insufficient. This gap underscores the need for a centralized system that combines smart attendance, lesson management, instructor tracking, and performance evaluation tailored to the operational and pedagogical needs of private music academies. Such a platform would allow for real-time monitoring, cross-branch coordination, and comprehensive assessment of student learning outcomes, addressing both administrative and instructional challenges in music education.

Research Gaps

Despite existing studies on educational and attendance management systems, several gaps remain, particularly in the context of private music academies:

Fragmented Data Management

Many institutions still rely on separate or manual systems for attendance, scheduling, and performance records, resulting in data inconsistency and operational inefficiencies.

Lack of Music-Specific Performance Tracking

Existing school management systems generally focus on academic grading and lack tools designed to monitor music-specific performance and student progress.

Absence of Multi-Branch Management Systems

Most available systems are designed for single-location institutions and do not support centralized management across multiple branches of a music academy.

Limited Integration of Smart Attendance Systems

Although QR code–based attendance solutions exist, they are seldom integrated into a unified web-based system tailored for private music schools.

CHAPTER 3

METHODOLOGY

Research Design

This study employed a developmental research design with a descriptive-evaluative approach. The developmental aspect focused on the systematic design, development, and implementation of the Web-Based Multi-Branch Music Academy Management System. The descriptive-evaluative approach was used to assess the system's quality, functionality, and usability after development.

The developmental approach is appropriate for this IT capstone since the study aimed to produce a functional software solution that addresses existing problems in attendance monitoring, lesson management, and performance tracking at Father and Sons Music Lessons.

The completed system was evaluated using the ISO/IEC 25010 Software Quality Model and the System Usability Scale (SUS) to ensure a structured and objective evaluation of software quality and usability.

Respondents and Sampling Technique

The respondents of the study consisted of three (3) individuals from Father and Sons Music Lessons who are directly involved in daily academy operations. These included the owner–super administrator and two (2) front-desk staff members.

A purposive sampling technique was employed, as the selected respondents possess direct knowledge and experience with student enrollment, attendance monitoring, lesson

scheduling, and report generation. Their feedback was used to evaluate the developed system based on ISO/IEC 25010 and SUS criteria.

Research Locale

The study was conducted at Father and Sons Music Lessons, a multi-branch music academy offering various music courses. The research covered the main branch and selected operational branches where administrative activities such as student enrollment, attendance recording, lesson scheduling, and performance monitoring are conducted.

The locale was selected to address challenges related to manual record-keeping, fragmented data, delayed report generation, and difficulty in monitoring attendance and performance across branches. Conducting the study in the actual operational environment ensured that the developed system aligned with real workflows and organizational needs.

Data Gathering Procedure

Permission was secured from the management of Father and Sons Music Lessons prior to data collection. Interviews and consultations were conducted with the owner and staff to identify current workflows, existing issues, and system requirements.

Direct observation of manual processes—such as student registration, attendance recording, lesson scheduling, and report preparation—was performed to determine inefficiencies. Existing documents, including attendance sheets and records, were also reviewed.

After system development, structured evaluation questionnaires based on ISO/IEC 25010 and the System Usability Scale (SUS) were administered to the respondents. The collected data were organized, analyzed, and interpreted to determine the system's quality and usability.

Research Instruments

The study employed **two standardized instruments** to ensure objective evaluation, supported by interviews, observation, and system testing:

1. **ISO/IEC 25010 Software Quality Questionnaire** – Adapted as a framework to assess the system across key quality characteristics: Functional Suitability, Performance Efficiency, Usability, Reliability, Security, Maintainability, and Compatibility. The framework guided evaluation during system testing and user feedback, ensuring the system met quality standards.
2. **System Usability Scale (SUS)** – A standard 10-item questionnaire used to measure perceived ease of use and learnability of the system. Responses were collected using a five-point scale (1 = Strongly Disagree to 5 = Strongly Agree) during system testing to complement the ISO evaluation.
3. **Interviews and Observation** – Structured interviews with the owner and staff, along with direct observation of daily operations, were conducted to gather qualitative insights on current manual processes, operational challenges, and user needs. Feedback from these activities helped validate system requirements and refine its functionality.

System Development Life Cycle (SDLC)

The study adopted the Modified Waterfall Model with Iterative User Feedback as the Software Development Life Cycle (SDLC). This model was chosen for its structured, sequential phases while still allowing continuous refinement through user feedback, ensuring that the final system met the operational needs of Father and Sons Music Lessons. The SDLC phases applied in this study included Planning, Requirements Gathering, Analysis & Design, Development, Testing & Evaluation, and Deployment.

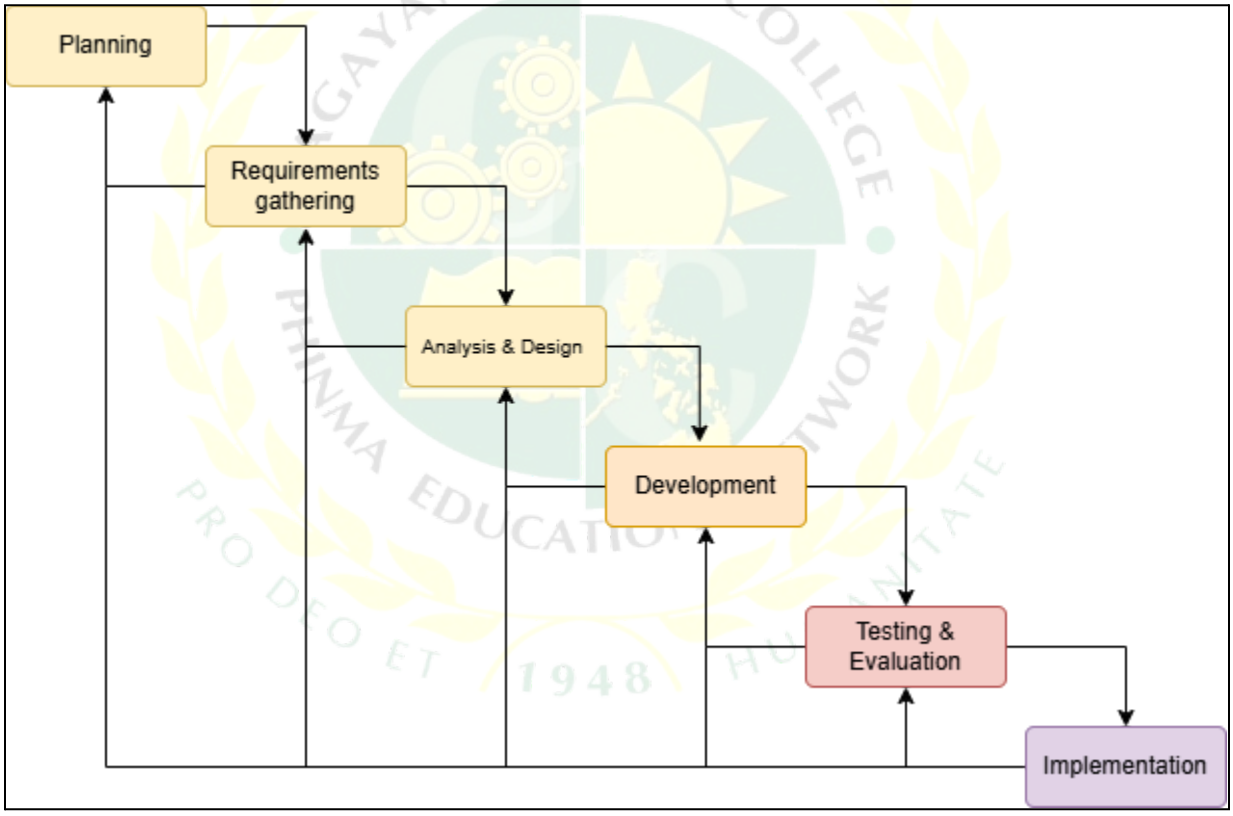


Figure. Modified Waterfall Model with Iterative Development

Planning Phase

The Planning Phase established the overall objectives, scope, and feasibility of the proposed Web-Based Multi-Branch Music Academy Management System for Father and Sons Music Lessons.

Discussions with the owner and staff identified key problems in the current manual processes, including paper-based student enrollment, attendance tracking, lesson scheduling, recital coordination, and report generation. These challenges resulted in data inconsistencies, delayed reports, and difficulties monitoring student progress across multiple branches.

Based on these discussions, the **system scope** was defined to include student and guardian management, lesson package enrollment, session scheduling, QR code attendance tracking, performance monitoring, recital management, and billing and ledger management. Delimitations were also set to ensure feasibility within the project timeline, such as excluding mobile application development and advanced multimedia features.

Requirements Gathering Phase

The Requirements Gathering Phase involved a detailed analysis of functional and non-functional requirements through interviews with the owner and front-desk staff, along with observation of daily operations at Father and Sons Music Lessons. Key functional requirements identified included a centralized multi-branch database, automated lesson scheduling, QR code-based attendance tracking, student performance monitoring, recital management, and report generation. Non-functional requirements focused on usability for staff and parents, reliability, data security, real-time access across branches, and consistent system performance during peak

enrollment periods. This phase ensured that the proposed system would address operational challenges, improve workflow efficiency, and meet the actual needs of its users.

Analysis and Design Phase

During the Analysis and Design Phase, the gathered requirements were analyzed and translated into system models and architectural designs. The researchers created Entity-Relationship Diagrams (ERD) to design the database, ensuring that student records, attendance, lesson schedules, and performance data are properly structured and synchronized across modules. Use case diagrams were developed to define the role-based access control (RBAC), specifying permissions and workflows for administrators, instructors, and staff according to their responsibilities. The system was designed as a centralized web-based platform, allowing seamless access across multiple branches. User interfaces were created to prioritize usability and efficient navigation, while security measures, including login authentication and access restrictions, were incorporated to protect sensitive information. This phase served as the technical blueprint guiding the system's development and implementation.

Developmental Phase

This stage followed a thorough comprehension of the system requirements and details. Following completion of the design illustrations, this was the actual construction process. The implementation phase involved the concrete realization of the planned system architecture and modules.

The system was built using the traditional three-tier (3-Tier) architectural pattern, which separates concerns into the Presentation, Application, and Data layers. This structure promotes

maintainability, scalability, and security. The Presentation Layer was developed using HTML5, CSS, and the Bootstrap 5 framework to create a responsive and user-friendly interface, with Axios.js handling asynchronous communication with the backend. The Application (Business Logic) Layer was implemented using PHP with the PDO (PHP Data Objects) extension to process requests, execute business rules, and serve a RESTful API. The Data Layer utilized a MySQL relational database to securely store and manage all system data, including student and guardian records, lesson packages, enrollment details, session schedules, instructor information, recital records, and payment transactions.

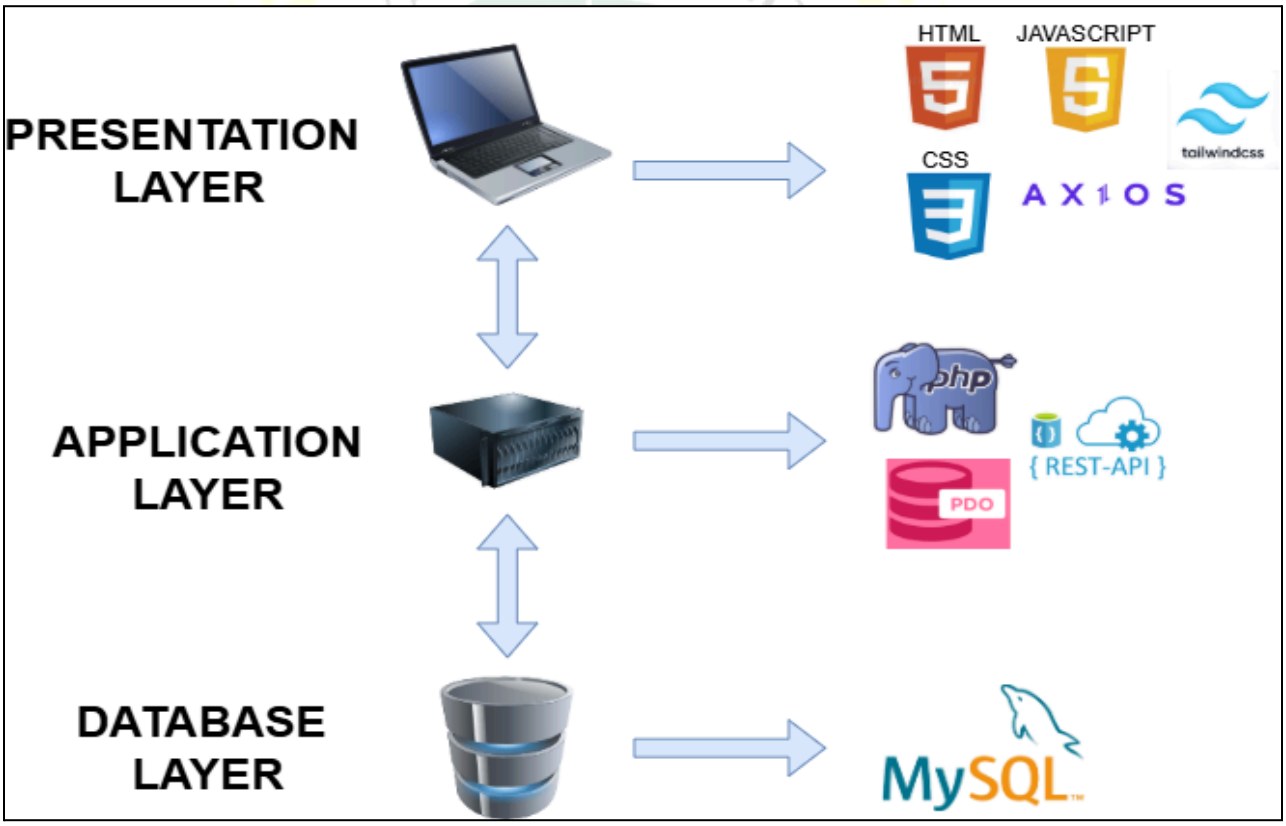


Figure 3.2: 3-Tier Architectural Design of the System

Testing and Evaluation Phase

In the Testing and Evaluation Phase, the Web-Based Multi-Branch Music Academy Management System for Father and Sons Music Lessons was assessed to ensure it met the

operational needs of the academy. The system was tested across all modules, including student enrollment, lesson and recital scheduling, smart attendance tracking using QR codes, performance management, instrument tracking, and payment processing. Specific test cases verified the accuracy of calculated revenues, proper recording of attendance and performance scores, and the timely triggering of automated reminders for lessons, recitals, and payments.

Scenario-based testing simulated real-world user interactions. For instance, administrators logging in were redirected to dashboards showing key metrics such as total enrolled students per branch, upcoming recitals, and pending tuition payments. Incorrect student login attempts prompted clear error messages and account recovery options, ensuring the system responded appropriately in both successful and failed scenarios.

Evaluation involved three respondents: the owner-superadmin, responsible for overseeing technical operations, and two front-desk staff members handling daily administrative tasks. Their direct involvement with existing manual processes ensured relevant and accurate feedback. The system was evaluated using two standardized instruments:

1. **ISO/IEC 25010 Software Quality Model**, assessing functional suitability, performance efficiency, usability, reliability, security, maintainability, and portability, with responses recorded on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree).

2. **System Usability Scale (SUS)**, measuring overall user satisfaction and perceived ease of use through ten statements rated 1 to 5, where higher scores indicate better usability.

Feedback from these tests informed system refinements to ensure it was reliable, user-friendly, and fully aligned with the academy's requirements.

Deployment Phase

The Deployment Phase was the final stage of the system development for the Web-Based Multi-Branch Music Academy Management System for Father and Sons Music Lessons with Smart Attendance and Performance Management. After thorough testing and evaluation, the system was configured on a web server and verified to ensure all modules—including student enrollment, lesson and recital scheduling, attendance tracking, performance management, and tuition monitoring—were fully functional and accessible to all users.

During this phase, the owner-superadmin and two front-desk staff members received hands-on training on the administrative, instructional, and reporting features of the system. Students and parents were oriented on how to access lesson schedules, monitor attendance, track progress, and view recital participation online. This phase ensured that the system effectively replaced manual processes, improved data accuracy, reduced administrative workload, and provided a centralized, efficient, and user-friendly platform for the academy's operations.

Tools, Technologies, and Technical Environment

The system was developed using the following tools and technologies:

Programming Languages: PHP, JavaScript

Frameworks/Libraries: Bootstrap, Axios.js

Database Management System: MySQL

Development Tools: Visual Studio Code, XAMPP

Operating System: Windows 11

Web Browser: Google Chrome

These tools and technologies were chosen for their compatibility, reliability, and community support. They allowed the creation of a secure, maintainable, and scalable management system suitable for the music academy environment.

Ethical Considerations

Ethical standards were strictly observed throughout the conduct of the study. Informed consent was obtained from all respondents prior to their participation. Participation was voluntary, and respondents were informed of their right to withdraw from the study at any time without penalty.

All data collected were treated with confidentiality and were used solely for academic purposes. Personal information was protected in compliance with the Data Privacy Act of 2012 (Republic Act No. 10173).

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